

The effect of curcumin and placebo on human gall-bladder function: an ultrasound study.

BACKGROUND: The extract of medicinal plants containing curcumin is traditionally believed to have a positive contraction effect on the human gall-bladder. **AIMS:** To compare the effect of 20 mg curcumin or placebo on the gall-bladder volume of healthy volunteers. **METHODS:** A randomized, double blind and crossover design study was carried out in 12 healthy volunteers (seven males and five females). Ultrasonography examination was carried out serially to measure the gall-bladder volume. The data obtained was analysed by paired Student's t-test. **RESULTS:** The fasting gall-bladder volumes of 15.74 +/- 4.29 mL on curcumin and 15.98 +/- 4.08 mL on placebo were similar ($P > 0.20$). The gall-bladder volume was reduced within the period after curcumin administration. The percentage of gall-bladder volume reduction at 0.5, 1.0, 1.5 and 2.0 h after 20 mg curcumin administration were 11.8 +/- 6.9, 16.8 +/- 7.4, 22.0 +/- 8.5 and 29.3 +/- 8.3%, respectively, which was statistically significant compared to placebo. **CONCLUSION:** On the basis of the present findings, it appears that curcumin induces contraction of the human gall-bladder.